

Kinesiology
KINESIOL 3E03
Neural Control of Human Movement
Winter 2027



SCIENCE

Instructor Information



Claire Tuckey
Email: tuckeyc@mcmaster.ca
Office Hours:
By appointment

Course Information

Lectures: See Mosaic

Labs: N/A

Tutorials: N/A

Course Dates: 01/04/2027 - 04/07/2027

Units: 3.00

Course Delivery Mode: In Person

Course Description: Neuromuscular control underlying human movement. Topics include basic neurophysiology, mechanisms of sensation, reflexes, voluntary movement and theories of motor control with special reference to brain function. Lectures (three hours); one term Prerequisite(s): KINESIOL 2MC3 (or KINESIOL 1E03) and registration in Level III or above of an Honours Kinesiology program
Cross-listed: LIFESCI 3K03

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)

- [Open Education Resources](#)

Course Learning Outcomes

- Appraise scientific literature at an introductory level
- Apply critical thinking skills to problems of human movement
- Review foundational concepts of neuroscience with a focus on human movement
- Describe central and peripheral neurological components involved in the sensory input and regulation of human movement
- Compare and contrast the role of central nervous system structures (spinal cord, thalamus, cerebellum, basal ganglia, cerebral cortex) in human movement
- Identify the importance of reflexes and central pattern generators in human movement
- Distinguish various movement pathologies and their neurological origin
- Identify new trends in the science and technology of human movement

Required Materials and Texts

Please sign in with your MacID [here](#) to view your booklist

Materials will be available on A2L

Course Fees

This term, we will be using Kritik360 (www.kritik.io/kritik360), a peer-to-peer learning and evaluation platform. Kritik360 puts you, the student, at the center of your learning. Kritik is focused on an important goal: to help you in developing critical thinking and communication skills, which in turn set you up for success both in and out of the classroom. **There will be a fee of \$29.00 CAD to use this software.** No need to have this downloaded prior to the start of the course, more instructions will be given in class.

Materials & Fees

Required Materials/ Resources

Students will be required to purchase and sign up for a peer-to-peer evaluation platform called “Kritik360”. This will be required to complete assignments, but students do not need to purchase this before the beginning of the course. Details on what it is and how to use it will be discussed in the course.

Kritik360

This semester, we will be using Kritik360 (www.kritik.io/kritik360), a peer-to-peer learning and evaluation platform. Kritik360 puts you, the student, at the center of your learning. Kritik is focused on an important goal: to help you in developing critical thinking and communication skills, which in turn set you up for success both in and out of the classroom.

Each Kritik360 activity has 3 unique stages:

- **Stage 1: Create** → Follow the instructions, read the provided rubric and create a submission
- **Stage 2: Evaluate** → Anonymously score your peers based on the rubric, and provide written comments
- **Stage 3: Feedback** → Provide your peer evaluators anonymous feedback in relation to how motivational and critical their written comments were

Grading

When you participate in Kritik360 activities, you will receive 3 scores: Creation score, Evaluation score, and Feedback score. The weighted average of these three scores will determine your Overall score for that activity. The Kritik360 Grading Scheme can be found in the course info section.

Registration/Support

Within the first week of the term, an email invitation will be sent to your McMaster email account that contains information on how to register for a Kritik360 account and enroll in the course.

- **Cost:** The subscription for this class is **\$29**. Once you create your account, you will be asked to make a payment to gain access to the course activities.

How to get help: If you have any questions about Kritik360, please use their live chat. A human agent will respond promptly within a few minutes from 9am-5pm eastern time

Monday to Friday. Outside of these hours, you'll receive a reply that they will be back the next business day - but don't worry! They monitor around the clock and will still respond to urgent requests within a few hours. You can also visit <https://help.kritik.io> to view their help articles.

Class Format

In Person

Course Evaluation

Course Evaluation

1. Topic Tests 40%
2. Research Paper Reflections Assignments 10%
3. Research Paper Discussion & Participation 10%
4. Final Exam 40%*

**This is the final assessment piece for the course. If this component is completed, you cannot withdraw from the course.*

Undergraduate Grading Scale

The grades breakdown for McMaster's 12-Point Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Course Schedule

Full course schedule will be outlined in lecture

Laboratory Overview, Operation, and Safety

N/A

Late Assignments

-10% per day late policy

Absences, Missed Work, Illness

See MSAF policy.

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Course Modification

The instructors and the university reserve the right to alter this outline if necessary

Generative AI: Use Prohibited

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it “shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source”. This includes work created by generative AI tools. Also stated in the policy is the following, “Contract Cheating is the act of “outsourcing of student work to third parties” (Lancaster & Clarke, 2016, p. 639) with or without payment.” Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

Course Specific Relief for Missed Academic Work Information

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Course Specific MSAF Policy

MSAF Test - Marks will be reweighted to the final exam, making it worth 50%

MSAF Research Paper Reflection - Marks will be reweighted to the other Research Paper Reflection, making it worth 10%

MSAFs ineligible during the Final Exam period.

Checking Grades & Regrade Policy

Checking Your Grades

Your marks will be recorded on Avenue. It is your responsibility to check that all grades entered into Avenue are recorded properly. All grade concerns and discrepancies must be reported to the Instructional Assistant within a week of receiving the grade.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#).

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism

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Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or

other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, Indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.